Quadratic equations and parabolas

1

a
- **i** Maximum
- **ii** Negative
- **iii** 2 solutions

b
- **i** Minimum
- **ii** Positive
- **iii** 0 solutions

c
- **i** Maximum
- **ii** Negative
- **iii** 1 solution

d
- **i** Minimum
- **ii** Positive
- **iii** 2 solutions

2 The equation has no real solution.

3

a 2 solutions **b** 1 solution **c** 0 solutions

The discriminant and equations

4

a
- **i** $\Delta = 0$
- **ii** 1

b
- **i** $\Delta = -76$
- **ii** 0

c
- **i** $\Delta = 1$
- **ii** 2

d
- **i** $\Delta = 16$
- **ii** 2

e
- **i** $\Delta = 0$
- **ii** 1

f
- **i** $\Delta = -23$
- **ii** 0

g
- **i** $\Delta = 40$
- **ii** 2

h
- **i** $\Delta = 100$
- **ii** 2

5

a $\Delta = 0$ **b** Rational

6

a
- **i** $\Delta = -144$
- **ii** Non-real roots.

b
- **i** $\Delta = 112$
- **ii** Real, unequal and irrational roots.

c
- **i** $\Delta = 0$
- **ii** Real, equal and rational roots.

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d

i $\Delta = 64$

ii Real, unequal and rational roots.

e

i $\Delta = 0$

ii Real, equal and rational roots.

f

i $\Delta = 65$

ii Real, unequal and irrational roots.

g

i $\Delta = -4$

ii Non-real roots.

h

i $\Delta = 3.75$

ii Real, unequal and irrational roots.

7

a $\Delta = 256$

b The roots are unequal and rational.

c $x = 12, x = -4$

8 There is only one real solution.

9 $b^2 - 4ac > 0$

10

a $9 + 8m$ **b** $16n^2 + 8$ **c** $45 - 4p$ **d** $16 + 4p^2$

e $12m$ **f** $\dfrac{-23n^2}{4}$

11 $a < \dfrac{9}{10}$

12 $m > -1$

13

a $m > -\dfrac{9}{20}$

b $m = 0$ must be eliminated as m is the coefficient of x^2 and by definition a quadratic equation must have an x^2 term.

14 $n = \pm 9$

15 $k = 1$

16 $m = -2, m = 4$

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17 **a**

i $64 - 8k$ ii $k = 8$ iii $k \leq 8$ iv $k > 8$

v $k < 8$

b

i $400 - 8k$ ii $k = 50$ iii $k \leq 50$ iv $k > 50$

v $k < 50$

c

i $52 - 12k$ ii $k = \frac{13}{3}$ iii $k \leq \frac{13}{3}$ iv $k > \frac{13}{3}$

v $k < \frac{13}{3}$

18

a $k > 74$ **b** $k = 75$

The discriminant and parabolas

19 None

20 One x-intercept

21

a

i $a > 0$ ii $\Delta > 0$

b

i $a < 0$ ii $\Delta = 0$

c

i $a < 0$ ii $\Delta < 0$

d

i $a > 0$ ii $\Delta < 0$

e

i $a > 0$ ii $\Delta < 0$

f

i $a > 0$ ii $\Delta > 0$

g

i $a < 0$ ii $\Delta = 0$

h

i $a < 0$ ii $\Delta > 0$

i

i $a > 0$ ii $\Delta = 0$

j

i $a < 0$ ii $\Delta < 0$

k

i $a < 0$ ii $\Delta = 0$

l

i $a < 0$ ii $\Delta > 0$

m

i $a < 0$ ii $\Delta < 0$

n

i $a > 0$ ii $\Delta = 0$

o

i $a < 0$ ii $\Delta > 0$

p

i $a > 0$ ii $\Delta < 0$

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22 **a** $m < -\frac{27}{4}$ **b** $m = -7$

23 $k = 16$

Application

24

a $0 < p < q$

b $x^2 + 2(p - q)x + p^2 - q^2 = 0$

c $\Delta = 8q(q - p)$

d Two real solutions as $q > p$, therefore the discriminant is positive.

e $x = 3p$

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